GG-C-00110a
September 1, 1971
Interim Revision of
Fed. Spec GG-C-110
July 16, 1958

INTERIM FEDERAL SPECIFICATION

CALIPER, X-RAY TECHNIQUE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-C-110.

1. SCOPE

1.1 This specification covers a caliper type device for measuring the thickness of body parts for determining X-Ray technique.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6525

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the priced indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

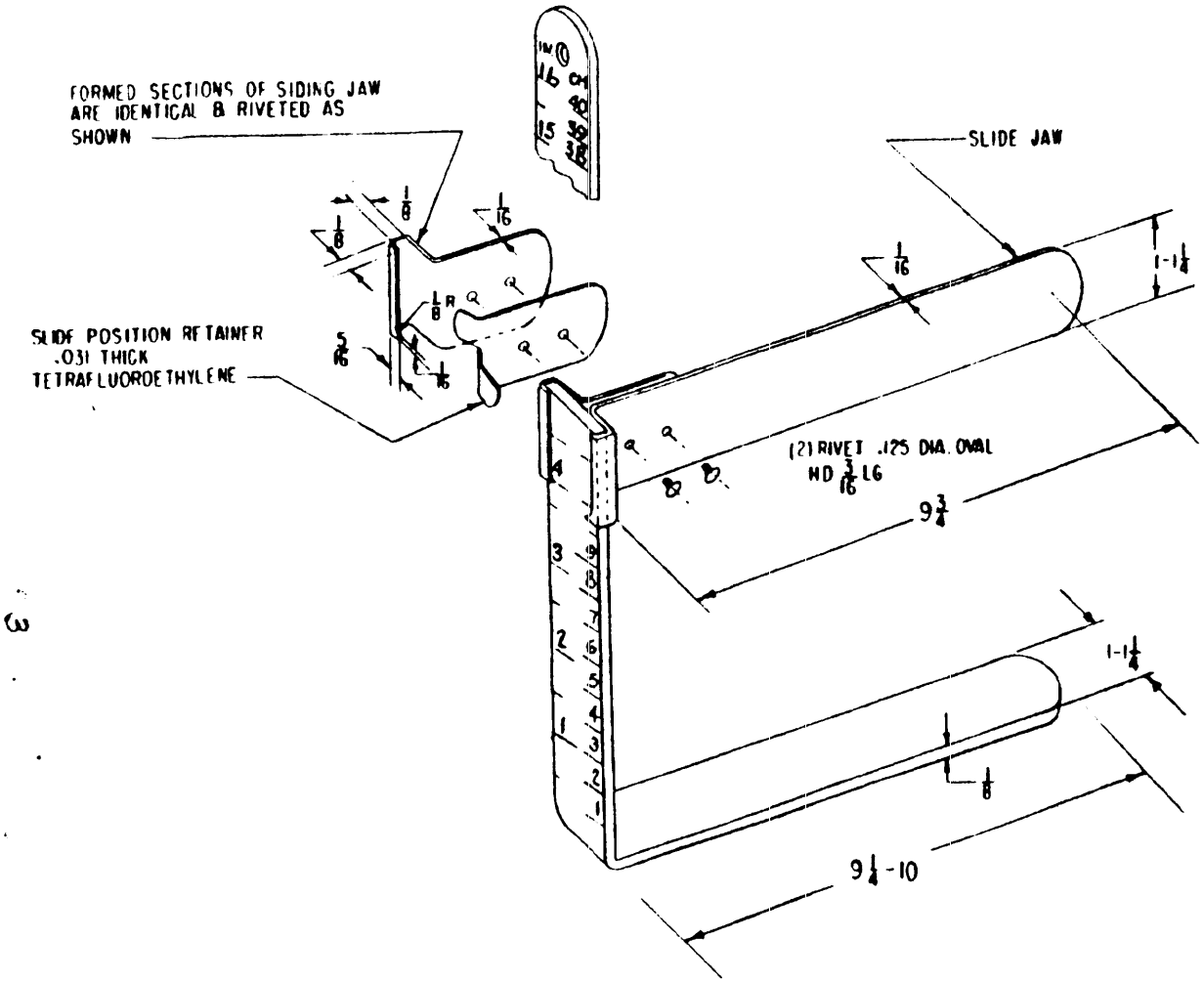
(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

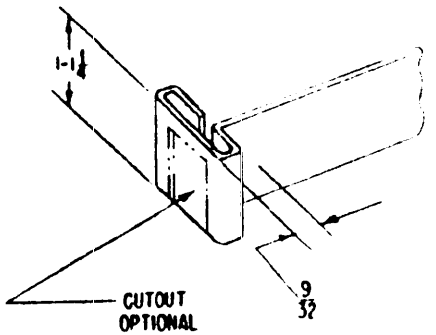
3. REQUIREMENTS

3.1 Material. Material used in the fabrication of the calipers shall be aluminum, alloy 6061-T6. The sliding jaw shall be 0.064 inch thick, No. 14 Brown and Sharpe (B and S) Gage. The stationary member shall be nominal 1/8 inch thick.

3.2 Construction. The caliper shall be designed and constructed as shown in figure 1.

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ALTERNATE DESIGN
(SLIDE JAW)

FIGURE 1
CALIPER X-RAY TECHNIQUE

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3.3 Performance. The sliding jaw shall move freely along the vertical scale but with sufficient tension to remain in any set position desired. When tested as specified in 4.4.1 the distance indicated on the scale shall be within 1/4 inch of the dimension of the test block.

3.4 Dimensions. Dimensions and tolerances shall be as shown on figure 1. The vertical scale shall be calibrated from 1 to 40 centimeters (cm.) in 1 cm. steps and from 1/2 to 16 inches in 1/2 inch steps. The zero point shall be at the base of the scale. The calibrations and numerals shall be stamped in the scale and the depressions shall be filled with black paint. The numerals shall be legible and of a permanent nature.

3.5 Finish. The caliper shall have a satin finish.

3.6 Identification marking. The name or registered trade-mark of the manufacturer, contract or purchase order number, Federal Stock Number, and the letters U. S. shall be stamped into the surface of the caliper.

3.7 Workmanship. Workmanship shall be first class throughout. Calipers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the aluminum, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-4	1.0
Minor defects	S-4	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 for an Inspection level of S-3 and AQL of 2.5.

4.3 Examination. The calipers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in Table II.

Table II

Categories	Defects*
Major	
101	Caliper not furnished complete with both component parts.
102	Slide device fails to slide smoothly and easily over vertical scale and retain any set position.
103	Scale graduations and numerals omitted in whole or part on vertical member.
104	Scale not located on proper surface of vertical member.
105	Scale range and graduations in English and metric units not as specified.

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Table II (cont'd.)

Categories	Defects*
Major	
106	Component parts not free of cracks or unspecified bends.
107	Component parts not free of burs and sharp edges.
108	Zero point of scale not coincident with top of base.
Minor	
201	Dimensions not within tolerances specified in figure 1.
202	1/4 inch hole near top of vertical scale omitted.
203	Identification marking omitted or not permanent and legible.
204	Finish not uniform.
205	Labels or printing on unit package not legible and free from defacing marks, scratches, and tears.

* Examination shall not be limited to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy test. The caliper shall be used to measure the thickness of a block of known dimension. The measurement shall be made with the portion of the caliper jaws farthest from the scale.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One caliper, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Required	36 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. The caliper shall be disassembled and wrapped in neutral tissue or other suitable anticorrosive material. The wrapped caliper shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676, or PPP-B-636, type CF, class domestic. Commercial colors are acceptable on unit packages and the closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. For measuring body parts for X-Ray techniques in medical facilities.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.4 This specification covers the following item shown in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6525-600-7900	CALIPER, X-RAY TECHNIQUE

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6525-1102

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R259

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION
QQ-C-00110a

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS OFFERORS

DO NOT CONDITION OR BASE YOUR BID OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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MAR 69

EDITION AUG 66 WILL BE USED UNTIL EXHAUSTED

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